

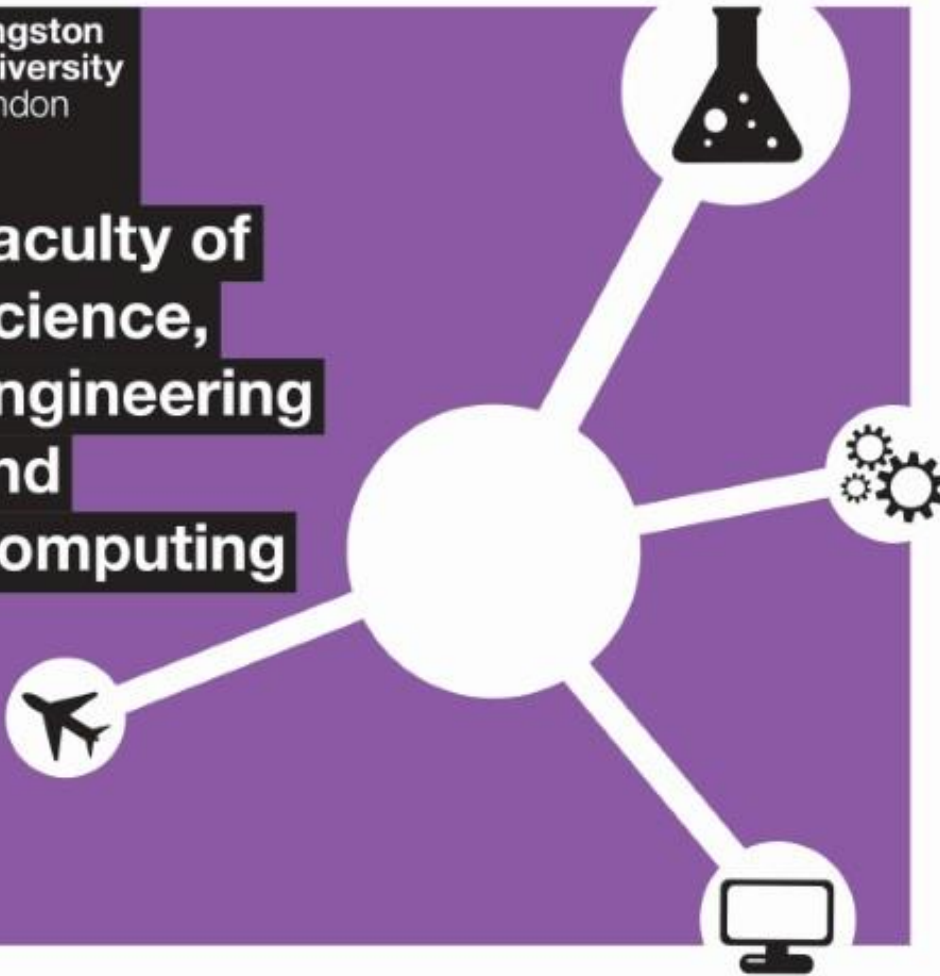
Class Props

- **None**

House Keeping

Kingston
University
London

Faculty of
Science,
Engineering
and
Computing



KEEP
CALM
AND
TURN PHONE
TO SILENT MODE

Mission Operations



AE7030 Space Mission Operations



Learning Objectives

- ✱ To gain an understanding of the ground segment
- ✱ To learn about the infrastructure and architecture of the ground segment
- ✱ To understand the operations of the ground segment
- ✱ To take on the role of a ground systems operator and as a team take an image of an asteroid

Space Mission Operations

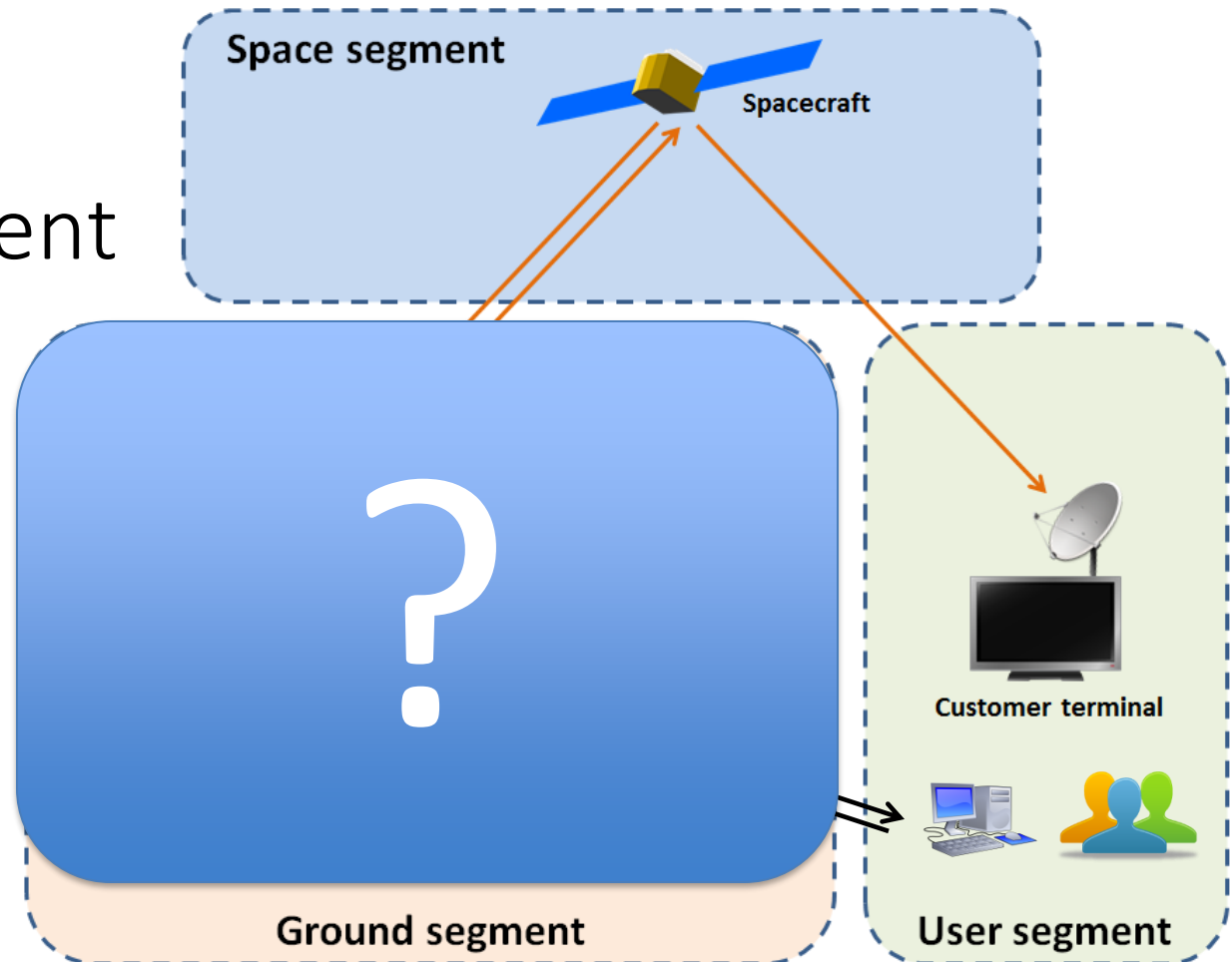
The Ground Segment

Space, Ground, and User Segment

Q1) What physical
infrastructure is part
of the ground
segment?

Q2) What operations
does the Ground
segment do?

(5 min)



Space Mission Operations

The Ground Segment: Infrastructure



Goonhilly, UK

Space Mission Operations

The Ground Segment: Infrastructure - ESTRACK

European Space Tracking network

A worldwide system of ground stations providing links between satellites in orbit and ESOC

9 stations in 6 countries each hosting one or more terminals comprising of an antenna and its signal processing equipment

3 Deep space antennas

Is able to up-link commands and down-link scientific data and spacecraft status information

Assists in ranging operations



Space Mission Operations

The Ground Segment: Infrastructure – ESTRACK Ground Stations



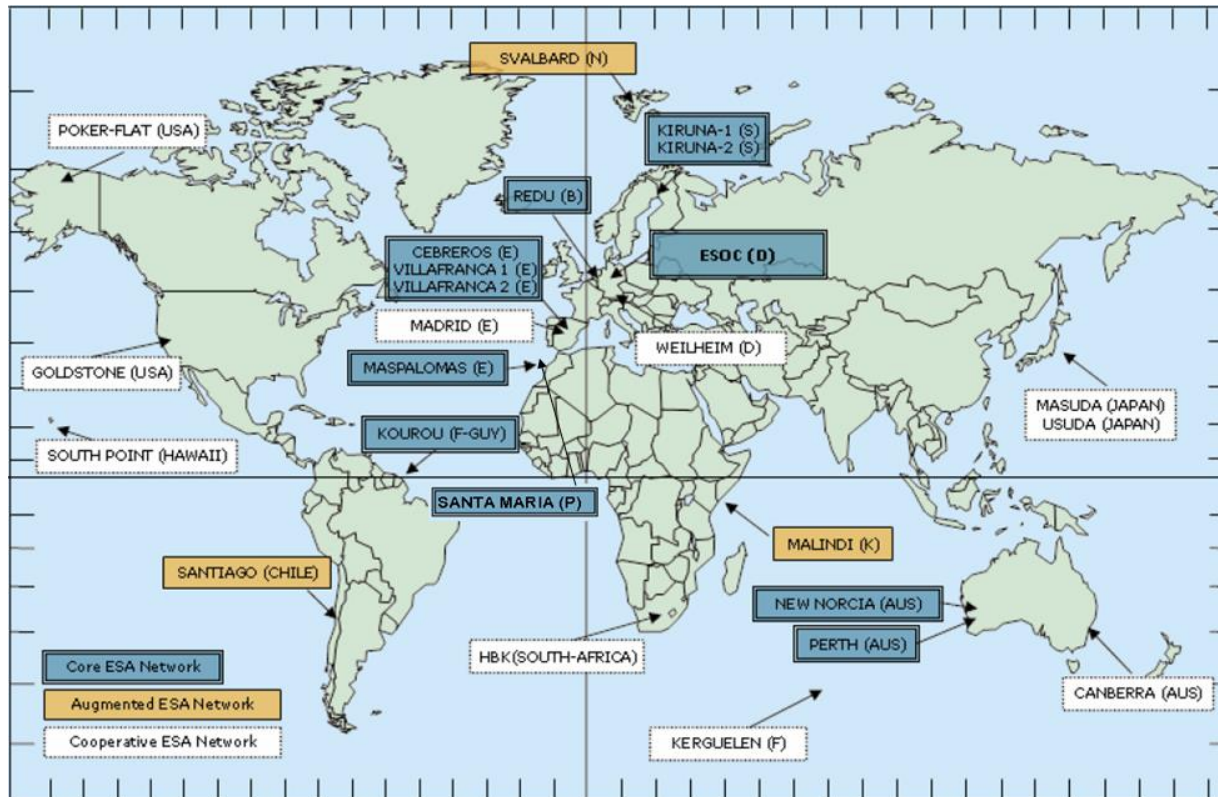
Cebreros



Santa Maria



Kourou



Kiruna



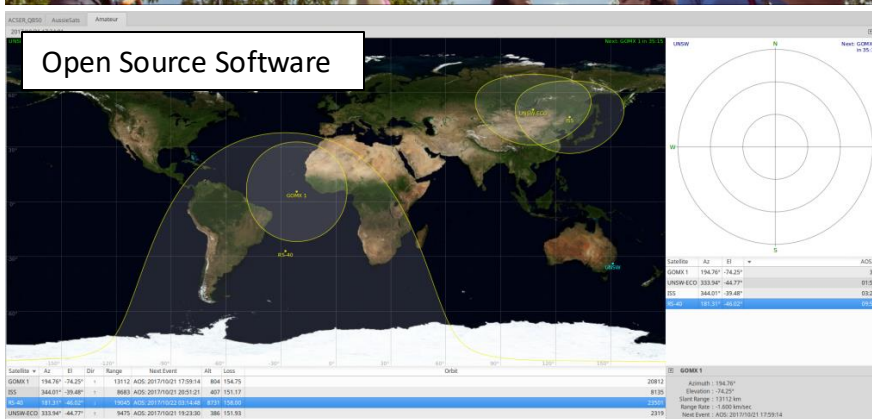
Redu



New Norcia

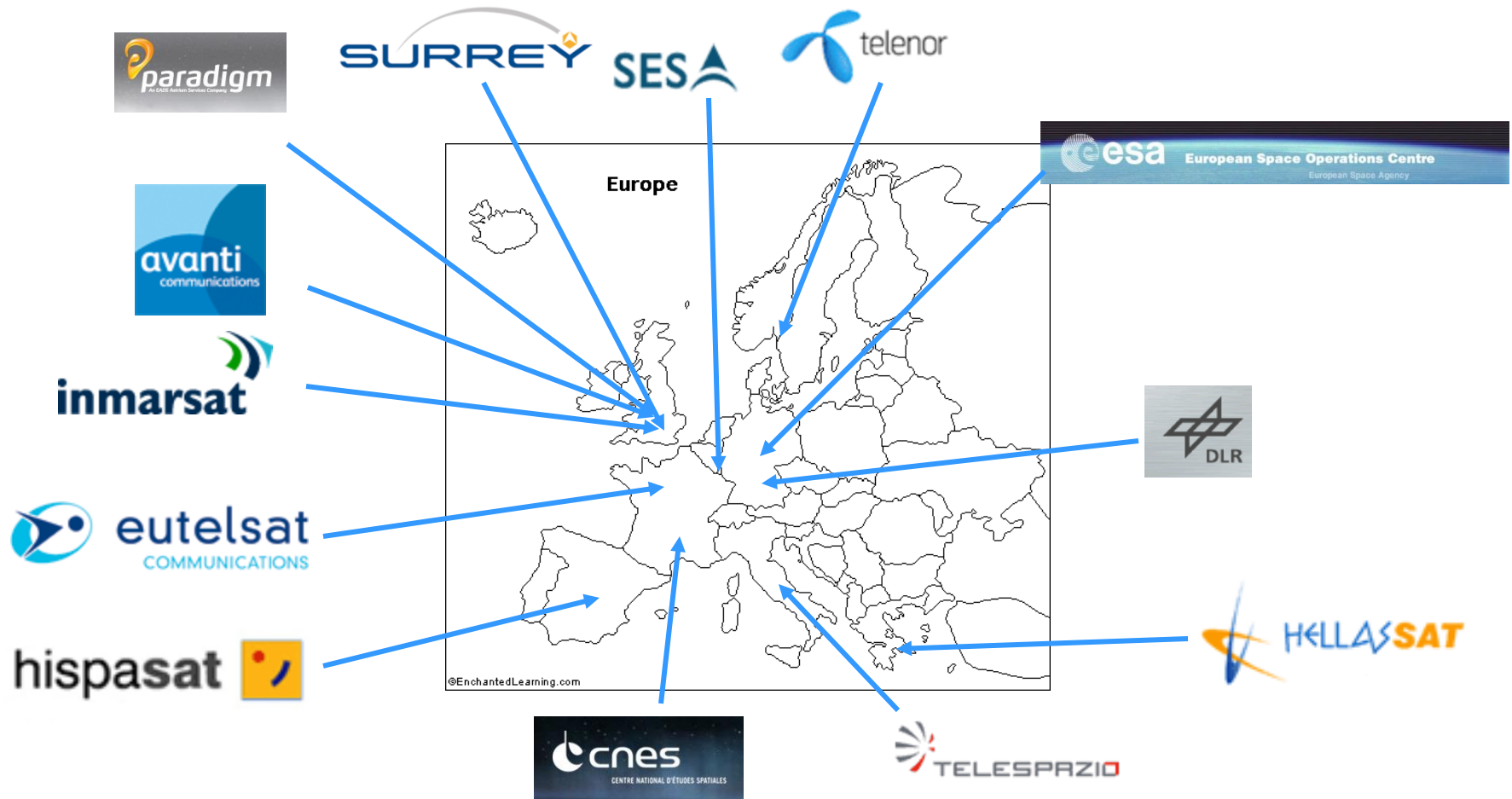
Space Mission Operations

The Ground Segment: Infrastructure - Alternatives



Space Mission Operations

The Ground Segment: Infrastructure – Mission Control Centres (around Europe)



Space Mission Operations

The Ground Segment: Infrastructure – Mission Control Centre (ESOC)



- ESOC is (ESA), located in
- ESOC is responsible for the operation of all ESA satellites and the ground segment
- ESOC works with
- ESOC has a number of numerous organisational



belongs to
all ESA
ance of
supported
national

Space Mission Operations

The Ground Segment: Infrastructure – Mission Control Centre (NASA)



Space Mission Operations

The Ground Segment: Infrastructure – Mission Control Centre (SSTL) - Layout

What common features can you spot with the ESA and NASA control room?

Ground Station Signal

UTC Clock

Ground Station Dish

Global Map & Orbit View

Viewing Window

Operators & Subsystem Leads

Operations Director

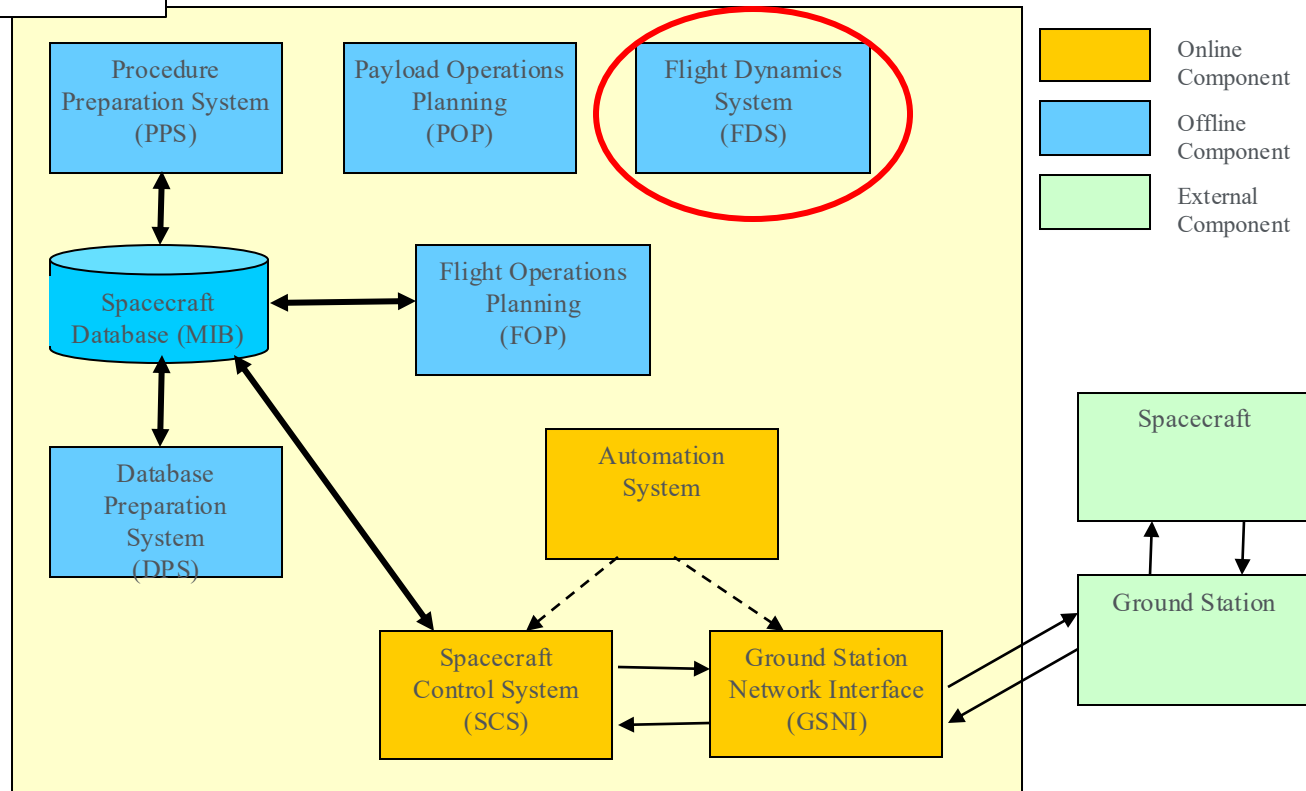


SSTL, UK

Space Mission Operations

The Ground Segment: Architecture

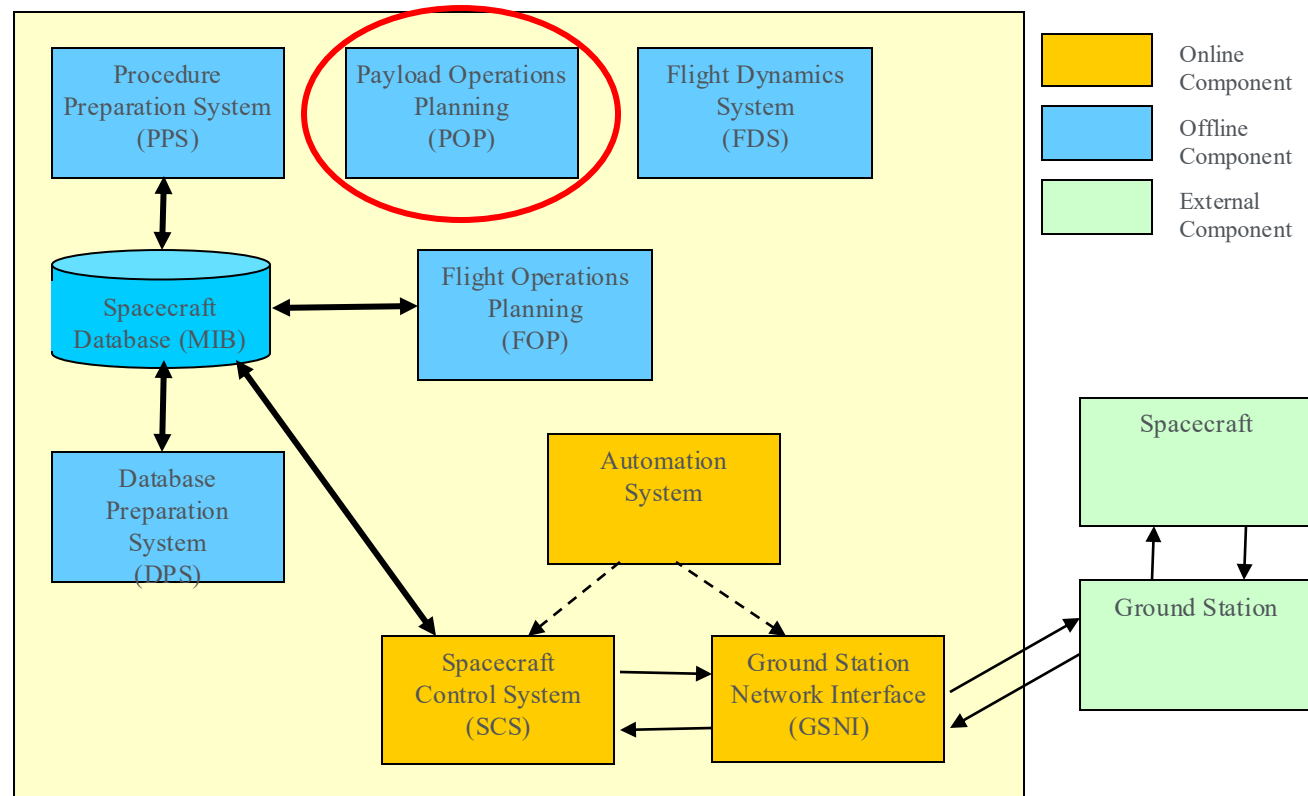
The **Flight Dynamics System** maintains and regularly updates a detailed spacecraft orbit model and generates information on the propagated orbit



Space Mission Operations

The Ground Segment: Architecture

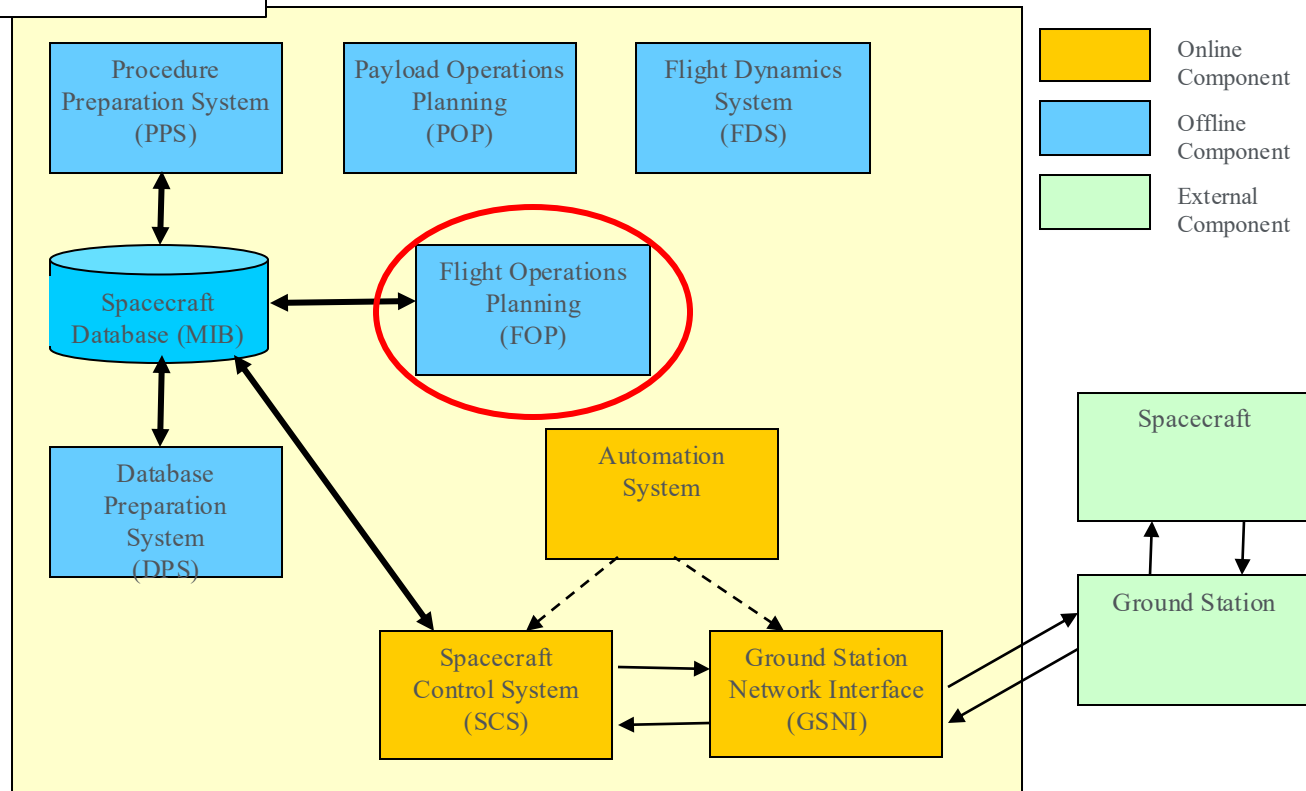
Payload Operations Planning processes requests for the acquisition of EO data to generate a payload operations plan



Space Mission Operations

The Ground Segment: Architecture

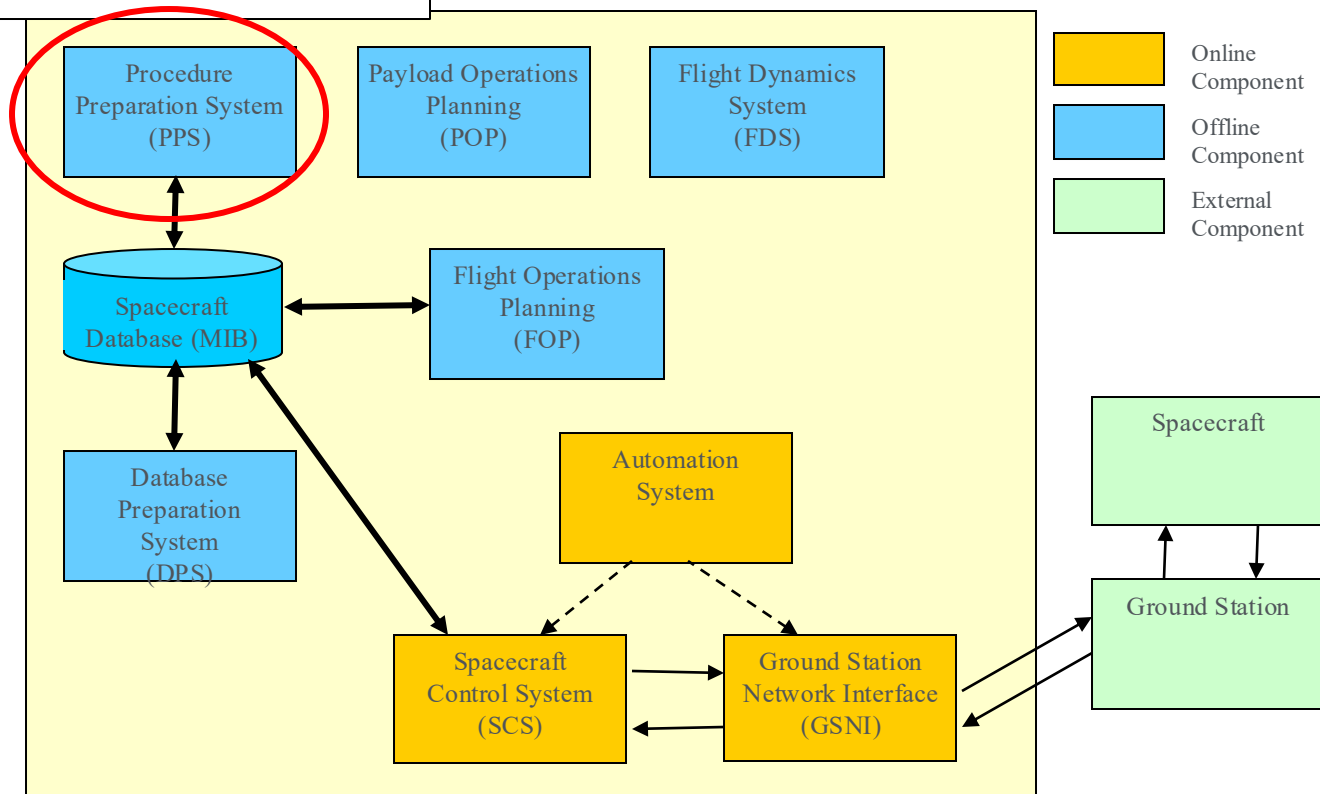
Flight Operations Planning consolidates information from Ground Stations, POP and FDS to produce an overall operations plan for both platform and payload



Space Mission Operations

The Ground Segment: Architecture

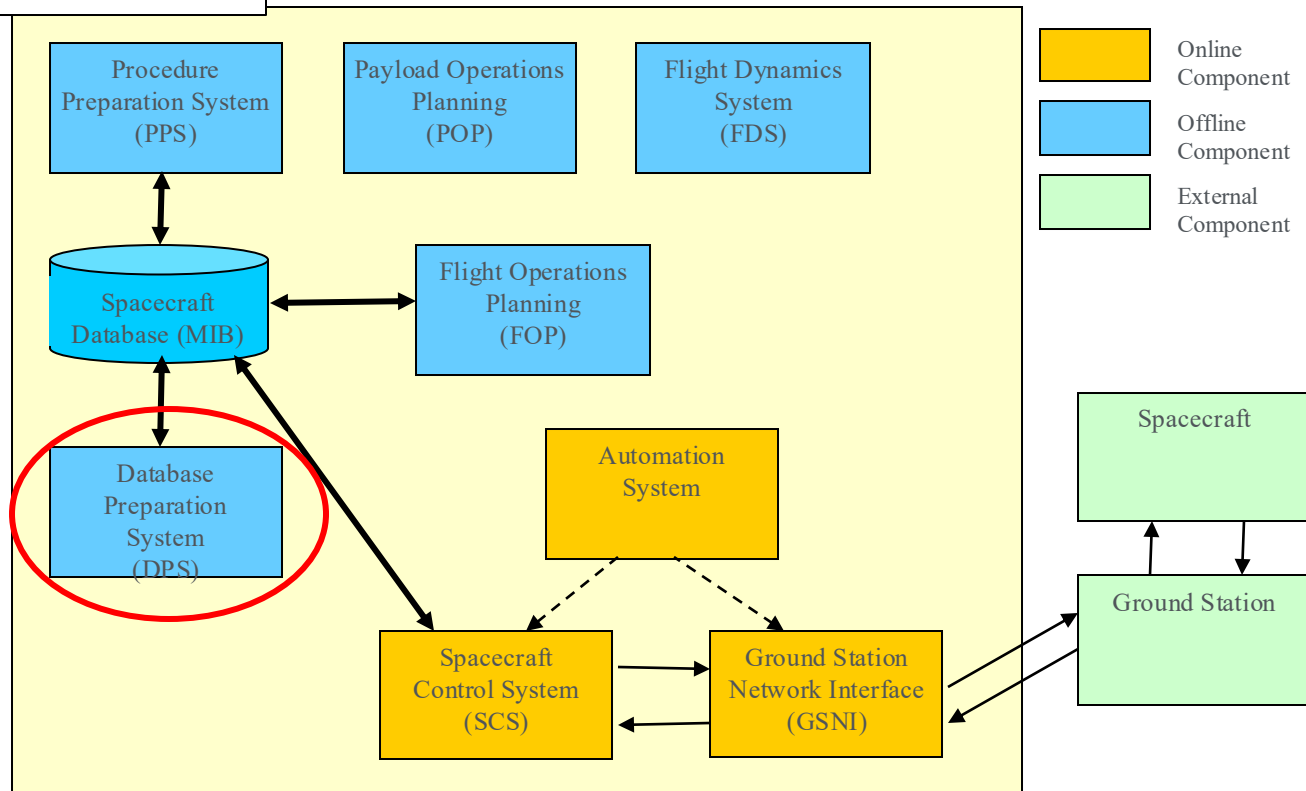
Procedure Preparation System is used to define and edit operational procedures. These are detailed sets of instructions which may be produced for either an Operator or the Automation System.



Space Mission Operations

The Ground Segment: Architecture

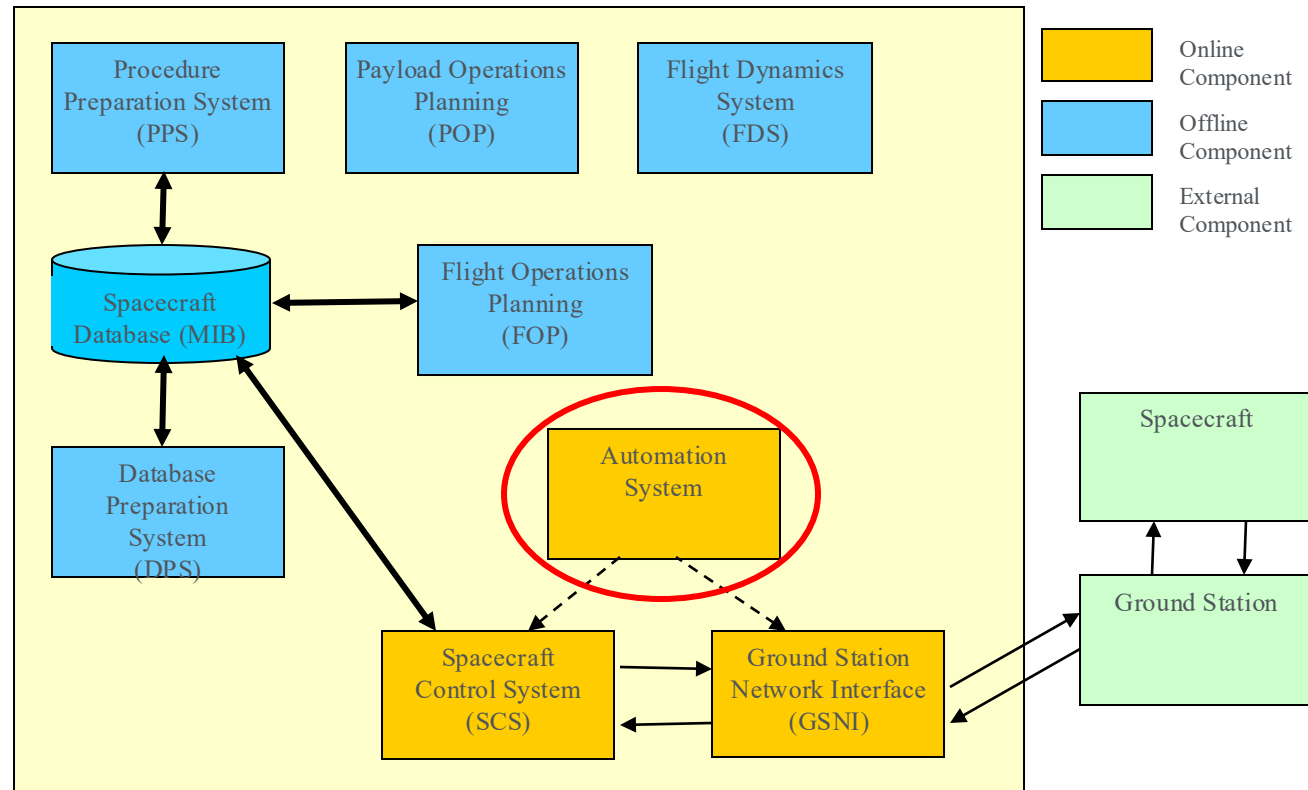
Database Preparation System is used to consolidate, update and validate the spacecraft database originally provided by the spacecraft manufacturer.



Space Mission Operations

The Ground Segment: Architecture

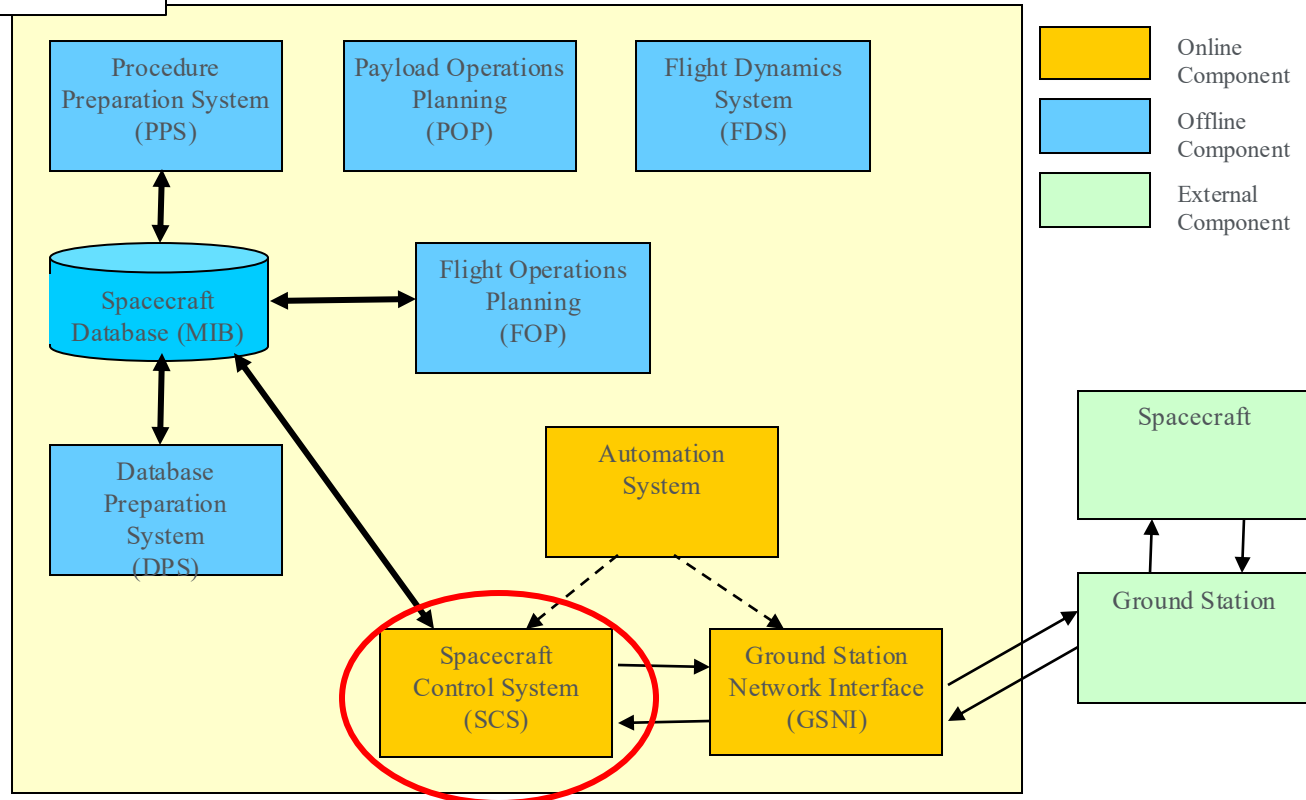
The **Automation System** “drives” the SCS and GSNI to perform their tasks without manual intervention



Space Mission Operations

The Ground Segment: Architecture

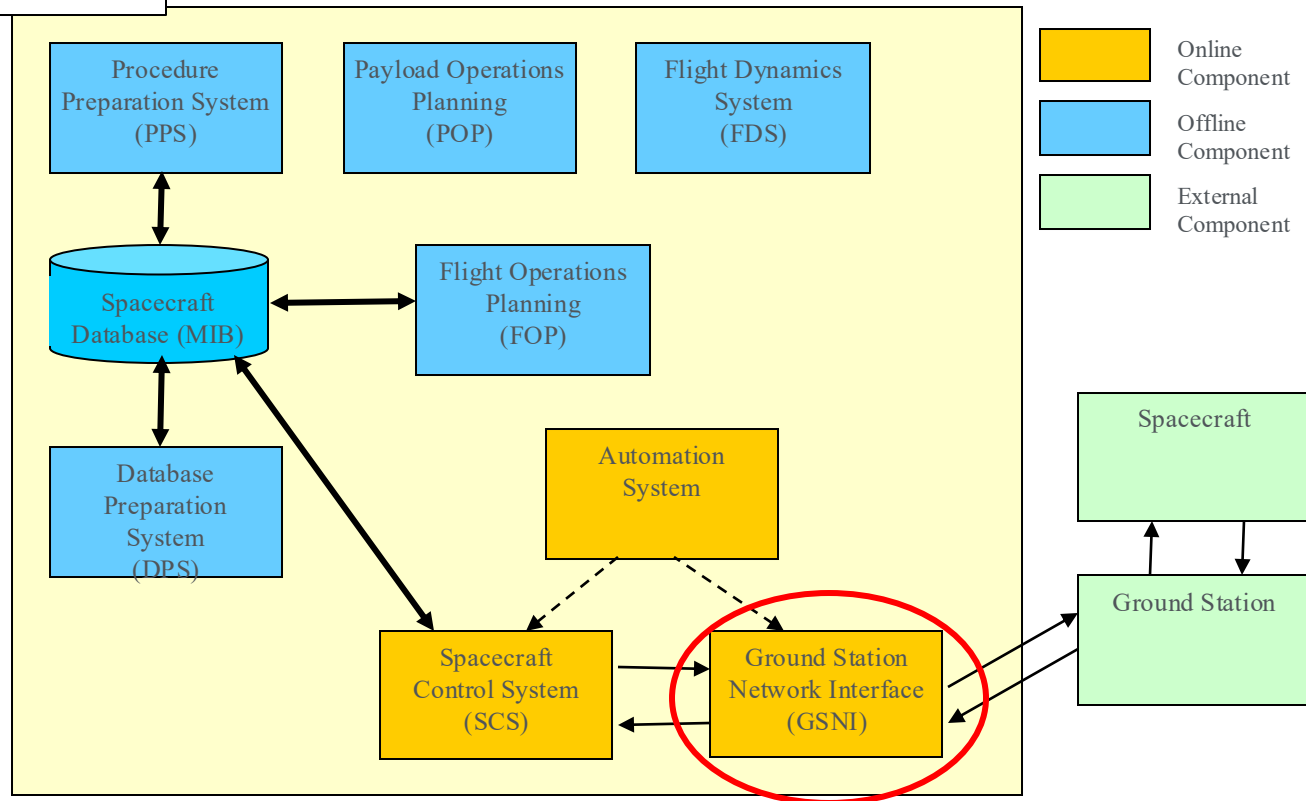
The **Spacecraft Control System** is the core component used to command the spacecraft and receive and process housekeeping telemetry.



Space Mission Operations

The Ground Segment: Architecture

The **Ground Station Network Interface** allows the SCS to connect to one of a number of available External Ground Stations.



Space Mission Operations

The Ground Segment: Infrastructure – Mission Control Centres (ESOC) – Mission Control Team



Space Mission Operations

The Ground Segment: Operations

Preparation for launch

- Ground segment preparation

- Spacecraft design support (Integration of S/C with Ground Segment, design for operability, requirements for mission performance)

Ground Stations for up- and down-link

- Inc. Communications networks

Operations Planning for Platform and Payload

Spacecraft Control & Monitoring (inc. Flight Dynamics)

- Health monitoring and problem solving

Payload Data / Management

- Signal generation (SatComs / Nav), Processing and Distribution

- Instrument / Data performance monitoring

- Archiving, Reprocessing and preservation (for Earth Observation)

End of Life

- Decommissioning of Spacecraft

- Long-term preservation and continuing improvement of data / reprocessing

- Lessons learned for next mission



Space Mission Operations

The Ground Segment: Operations

Rescue Operations

In rare cases, an unexpected event turns out to have a major impact on the ability to continue a mission as originally foreseen and it becomes necessary to embark upon a major recovery effort. Time is of the essence. Extensive rescue training and preparation should be undertaken before launch.

The [#MetOp](#)-C Flight Control A-Team



The [#MetOp](#)-C Flight Control B-Team

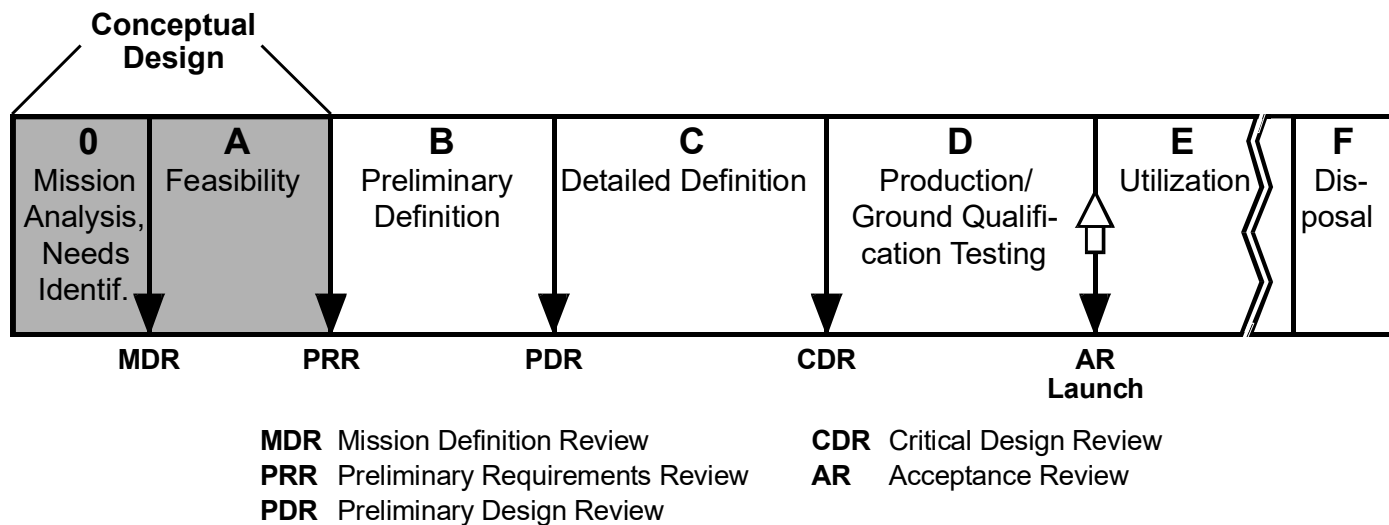


Train on Simulators, FlatSats, in mission control and training rooms

Project Life Cycle Overview

Operations activities should start very early in the lifecycle

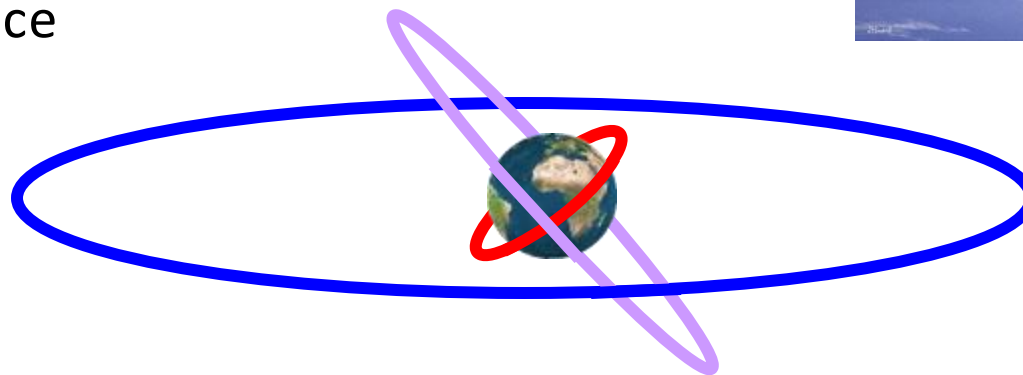
“Design For Operability” is vital



Orbits

Operations are strongly influenced by the satellite's orbit

- Low Earth Orbit (LEO) inc. Polar Orbit
- Medium Earth Orbit (MEO)
- Geostationary Orbit (GEO)
- Deep Space



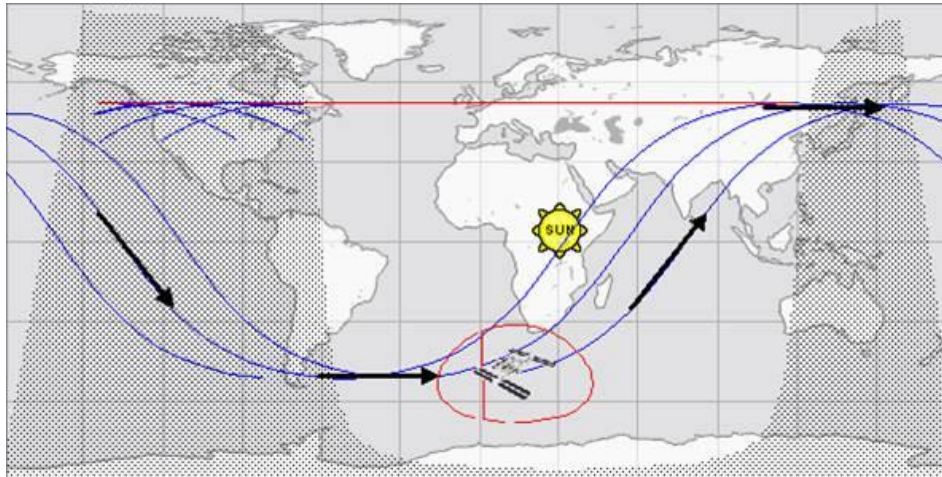
How would operations differ?

Space Mission Operations

Place-Washington-To-Satellite-TacSat: Access Summary Report (01 Mar 2019 10:12:00)

Asset-To-TacSat

Access	Start Time (UTCG)	Stop Time (UTCG)	Duration (sec)
-----	-----	-----	-----
1	1 Mar 2019 11:23:11.496	1 Mar 2019 11:27:10.326	239.830
2	1 Mar 2019 12:52:23.930	1 Mar 2019 13:08:24.159	961.229
Mean Duration			600.529



Pre-Launch Operations - What are the Ops Team doing?

- **Influences the Satellite design** – to ensure that the satellite provides operators with the right information and the right controls
- **Reviewing Documentation** – this is often the first port of call in an emergency and must be clear, complete and accurate
- **Undergoing Training** – Operations engineers need to learn about the spacecraft – and each other.
- **Configuring the Mission Control System.** E.g.:
 - Checking and testing all Telemetry & Telecommands
 - Preparing automated command sequences

Pre-Launch Operations - What are the Ops Team doing?

- **Reviewing/Writing/Testing Procedures** – need to be written and reviewed by subsystem experts and tested against the simulator
- **System Verification Tests (SVTs)** – Complete end-to-end test with the real spacecraft connected to the mission control centre
- **Launch Rehearsals** – simulation campaign

Launch and Early Orbit Phase (LEOP)

The first days / months of the mission from launch until the spacecraft is fully commissioned and all in-orbit testing completed

It includes:

- Final preparation for launch
- First contact with the spacecraft
- Monitoring/performing initialisation sequences
- Achieving the final operational orbit/trajectory
- Commissioning and configuring the spacecraft



Launch and Early Orbit Phase (LEOP)

What needs to be done?

- FIRST!
 - Communications & Power (battery)
- Switching on the satellites systems
 - Thruster network
 - Power Switches
 - Computer Start-up and Equipment warm-up
- Getting control of the satellite
 - Attitude Control
 - Computer control modes
- Getting to the right orbit, attitude (orientation) and deployment
- Setting up the payload for services (Commissioning)

Commissioning the Spacecraft

- In-Orbit Testing (IOT) is also performed which involves verifying and characterising the functionality of all spacecraft units. In order to minimise unnecessary commanding the usual sequence is:
 - Switch ON Redundant units
 - Switch OFF Redundant units and Switch ON Nominal Units

Routine Operations

Three types of activities:

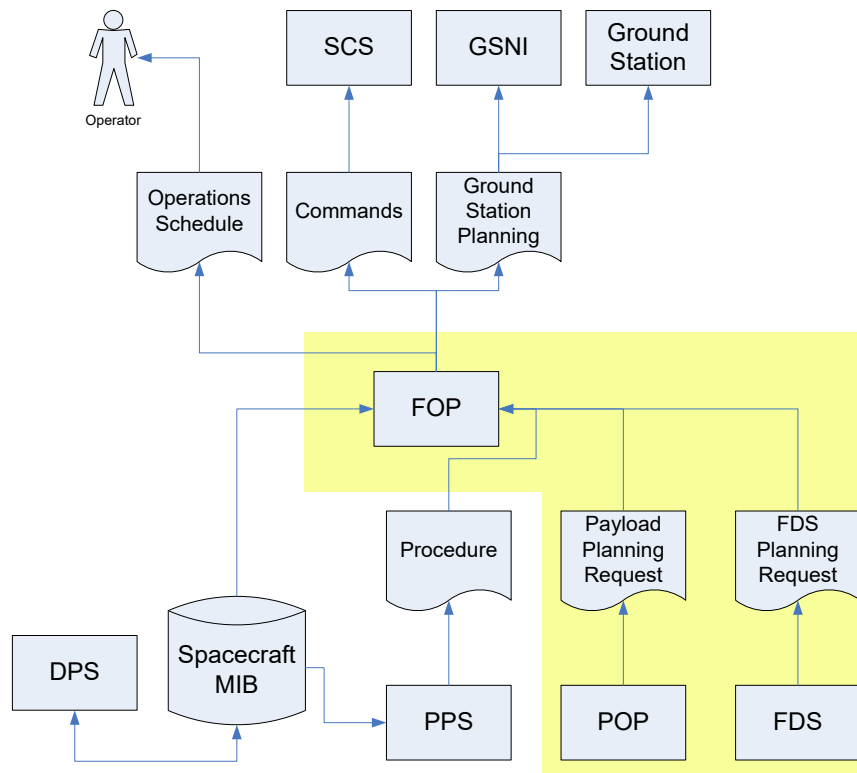
- **Preparation and Configuration** – performed on an *ad-hoc* basis
- **Operational Planning** - performed on a *daily* basis
- **Plan Execution** – performed on a *(near) continuous* basis

Preparation and Configuration

- Definition and maintenance of Flight Operations Procedures for M&C of satellite
- Definition and maintenance of Ground Operations Procedures for M&C of Ground Segment
- Maintenance and update of spacecraft database



Routine Operations

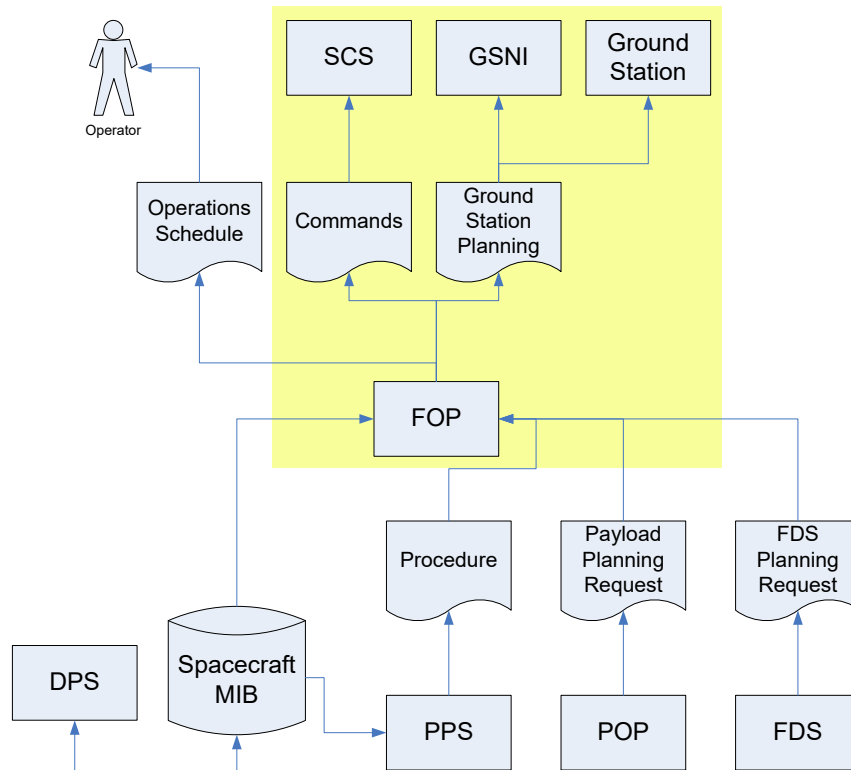


Operations Planning

These are manual activities which include:

- Generation of orbital events information
- Generation of payload planning requests
- Collating payload planning, FDS and Ground Station inputs into a master plan
- Producing command schedule for spacecraft
- Producing command schedule for Ground Segment
- Producing operator schedule

Routine Operations



Plan Execution

These are largely automated activities which include:

Regular pass activities:

- Establish satellite contact
- Uplink on-board schedule
- Downlink housekeeping telemetry
- Terminate satellite contact

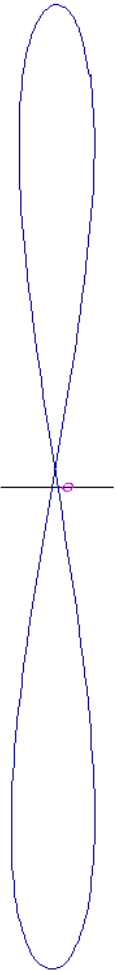
Routine monitoring of:

- Plan execution
- Satellite health status

Routine Operations

Geostationary Orbits

- Geosynchronous satellites orbit the Earth at the same rate as the Earth's rotation on its own axis. Hence, they maintain a Fixed Longitude position
- Geostationary satellites maintain a Fixed Longitude position AND a Fixed near-equatorial Latitude. This is achieved by controlling the spacecraft Inclination, i.e. keeping it at or near to Zero
- If the orbital inclination is not zero, the satellite will move in a figure-8 in the sky, every day



Routine Operations

Geostationary Orbits

- A TM/TC link can usually be maintained 24 Hours a Day
- Only one Ground Station is required (though back-ups may be available)
- Geostationary satellites slowly drift away from their 'perfect' position
- A longitude 'Box' (of say $\pm 0.1^\circ$) is defined. It is then necessary to perform a manoeuvre every week/month (as needed) to keep the spacecraft in that specified area of space (the 'box')

Routine Operations

Geostationary Orbits

- Manoeuvres are generally split into the following categories
 - **East / West Station Keeping** - these are the manoeuvres performed on all geosynchronous and geostationary satellites to control drifts in longitude. *Low fuel consumption.*
 - **North / South Station Keeping** - these are the manoeuvres performed geostationary satellites to control drifts in inclination. ***High fuel consumption.*** *Sun & Moon are the main cause of drift.*
 - **Eccentricity Correction** - to re- circularise the orbit. They are often combined with one of the above manoeuvre types. *Medium fuel consumption; Solar (light) pressure contributes to eccentricity*
 - **Attitude Manoeuvres** - these are the manoeuvres performed to dump momentum acquired through external torques (e.g. solar radiation pressure, gravitational torque etc.). *Low fuel consumption.*

Routine Operations

Geostationary Orbits

Pointing at the Earth –
The Sun & Moon can
cause problems...



Routine Operations

Geostationary Orbits

Pointing at the Earth
—
The Sun & Moon



Routine Operations

Eclipses in Geostationary Orbits

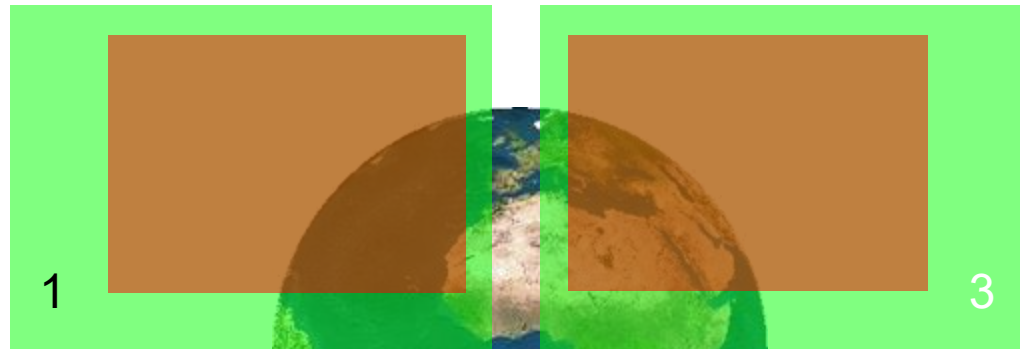
- For the majority of the year spacecraft in these orbits are never in shadow. However, during two short periods every year these orbits experience **Eclipse Seasons**
- This is due to the tilt of the Earth's axis with respect to the plane in which the Earth orbits the sun (the ecliptic)
- Eclipse seasons occur for a period of a few weeks around the Vernal (Spring) and Autumnal Equinoxes

Routine Operations

Geostationary Orbits

Sub Blinding & Eclipse around Equinox

Pre - Eclipse Season



Autumn Equinox



Post - Eclipse Season



4

4

Routine Operations

Low Earth Orbits

- Other types of Earth orbits generally present more operational complexities
- How would LEO differ?
- Eclipses many times per day in **Low Earth Orbit**
- Maintaining an RF link visibility of the satellite is much more complicated:
 - **Multiple Ground Stations** - in order to maintain full coverage a number of different ground stations around the globe are needed. This means accurate prediction of the visibility from each ground station as well as pointing of the antennas
 - **Link Loss** - where there are insufficient ideally placed ground stations there will be periodic loss of TM and TC
- Periodic link loss means that operations need to be planned very carefully

Routine Operations

Deep Space Missions

Deep Space Missions present a number of additional complex issues which vary greatly from mission to mission.

What sort of issues ?

- **Experimental Hardware** - specialist operations knowledge
- **Maintaining TC Link** - low signal strength, pointing accuracy
- **Orbit Determination** - modelling flight paths non trivial
- **Light Time** - takes minutes or hours to verify commanding
- **Solar Conjunctions** - Sun can obscure spacecraft for weeks
- **Ground Networks** - large (rare) dishes needed. Often shared
- **Non-Routine Operations** - some never enter 'routine' ops
- **Changing Environments** - these can vary greatly from near-solar, deep space and dusty near-comet environments

Routine Operations

Other Operations

- There are various routine operations tasks which can apply equally to all mission types. These can include:
 - **Thruster Efficiency** (Trend analysis) - Checking the efficiency of thrusters
 - **Tank Balancing** – moving fuel from one tank to another, to keep the spacecraft balanced
 - **Fuel Analysis** – Tracking the amount of fuel left in the tanks
 - **Inaccurate Modelling** - thermal modelling before launch may not be perfectly accurate and so some adjustment may be necessary during the early life of the spacecraft
 - **Replacement Heaters** - Keep the spacecraft warm when equipment is switched off
 - **Compensation Heaters** - Used to compensate when satellite insulation gets old and less effective
 - **Seasonal Changes** - see the following slide

De-Orbiting Operations

- These are also known as **Graveyard Manoeuvres**. This is the means by which an Earth-orbiting satellite is decommissioned at the end of its life and its orbital slot is vacated
- In the case of **geostationary satellites** the orbit is **boosted** from the (crowded) geostationary ring by about 300 km into a Graveyard Orbit. This is generally high enough to allow thousands of years before the spacecraft would return to Earth
- **Enough fuel** must be reserved in the tanks for the final manoeuvre

How do you measure how much fuel is in the tank?

De-Orbiting Operations

- Low Earth orbiting satellites can be decommissioned by performing re-entry manoeuvres
- Normally, they burn up in the atmosphere (some parts may hit the surface)



Space Mission Operations

Payload Dedicated Ground Segment



*Fucino,
Telespazio's
Teleport*

Receives and processes data from the spacecraft payload(s)

Communications -

- Routes data traffic to and from the satellite
- Manages network configuration and connects to billing systems

Earth Observation -

Payload Data Ground Segment (PDGS) receives “Raw” data from space segment and processes into useable data products, for end users (e.g. scientists, meteorologists, etc)

Satellite Navigation -

Galileo has a ‘Mission Segment’ which generates navigation information to be sent to and then broadcast from the spacecraft.